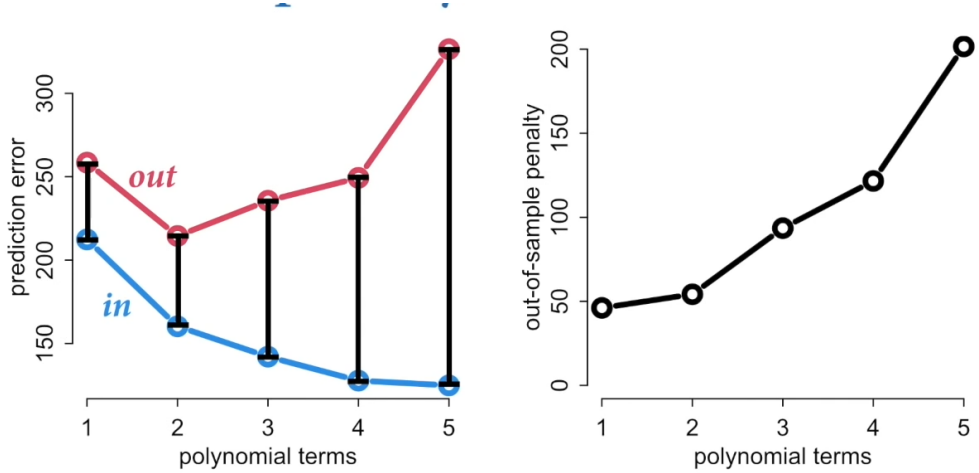
Lecture 20: Overfitting II

- Where we left it:



- If we increase model complexity, the discrepancy between In- and Out-samples error increases.
- The In-error always diminishes.
- The Out-error seems to have a favourite model.
- For N points, LOO-CV is *very* expensive.
- What if we could compute the penalty from a single model fit? Good news: PSIS (importance sampling) and WAIC (information criteria).

Ockham's razor

Nunca ponenda est pluralitas sine necesitate

(Plurality should never be posited without necessity)

- aka Models with fewer assumptions are to be preferred.
- In practice, we have to choose between models that differ both in accuracy and simplicity. The razor is not really a useful guidance for this trade off. The tools aforementioned (PSIS or WAIC) do.
- We want to avoid both *overfitting*, i.e. learning too much from the data and *underfitting*, i.e. learning too little from the data. Both would lead to poor predictions.
- Don't forget about *confounding*, a third problem. Yet, some confounded models can produce better predictions than models than correctly account for a causal relationship!?!
- Before presenting the complementary regularisation and information criteria paradigms, I will spend some time presenting *information theory*.

Fit to the data

Start of the odyssey

- Problem: fit to sample always (multi-level models can be counter-examples) improves as we add parameters.
- Dangers of "stargazing": selecting variables with low p-values (aka 'lots of stars'). P-values are not designed to cope with over-/under-fitting.
- An example of fit: the R^2 defined as:

$$R^2 = \frac{\text{var}(\text{outcome}) - \text{var}(\text{residuals})}{\text{var}(\text{outcome})} = 1 - \frac{\text{var}(\text{residuals})}{\text{var}(\text{outcome})}$$

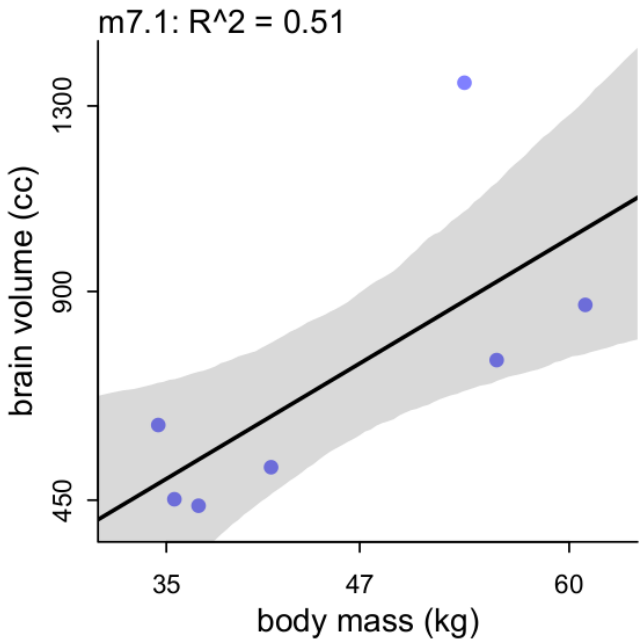
  The proportion of variance explained.
- I am sure you know how the R^2 behaves...

Fit to the data

R^2 hominin brains example

Simplest model

$$
\begin{aligned}
b_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha + \beta m_i \\
\alpha &\sim \text{Normal}(5, 1) \\
\beta &\sim \text{Normal}(0, 10) \\
\sigma &\sim \text{Log} - \text{Normal}(0, 1)
\end{aligned}
$$



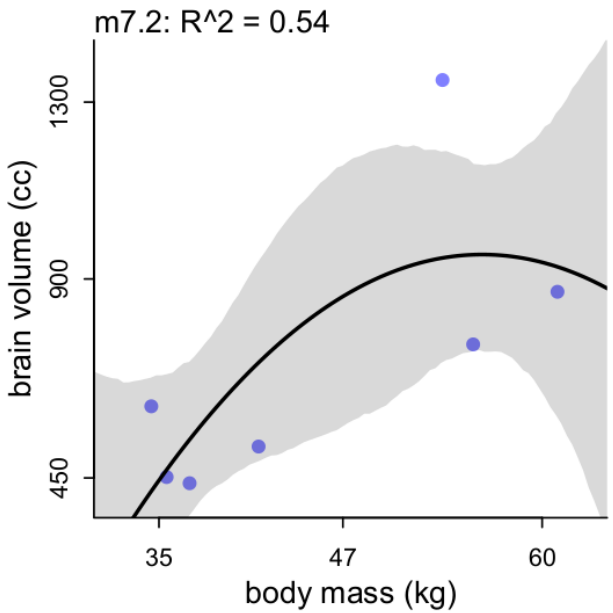
Fit to the data

R^2 hominin brains example

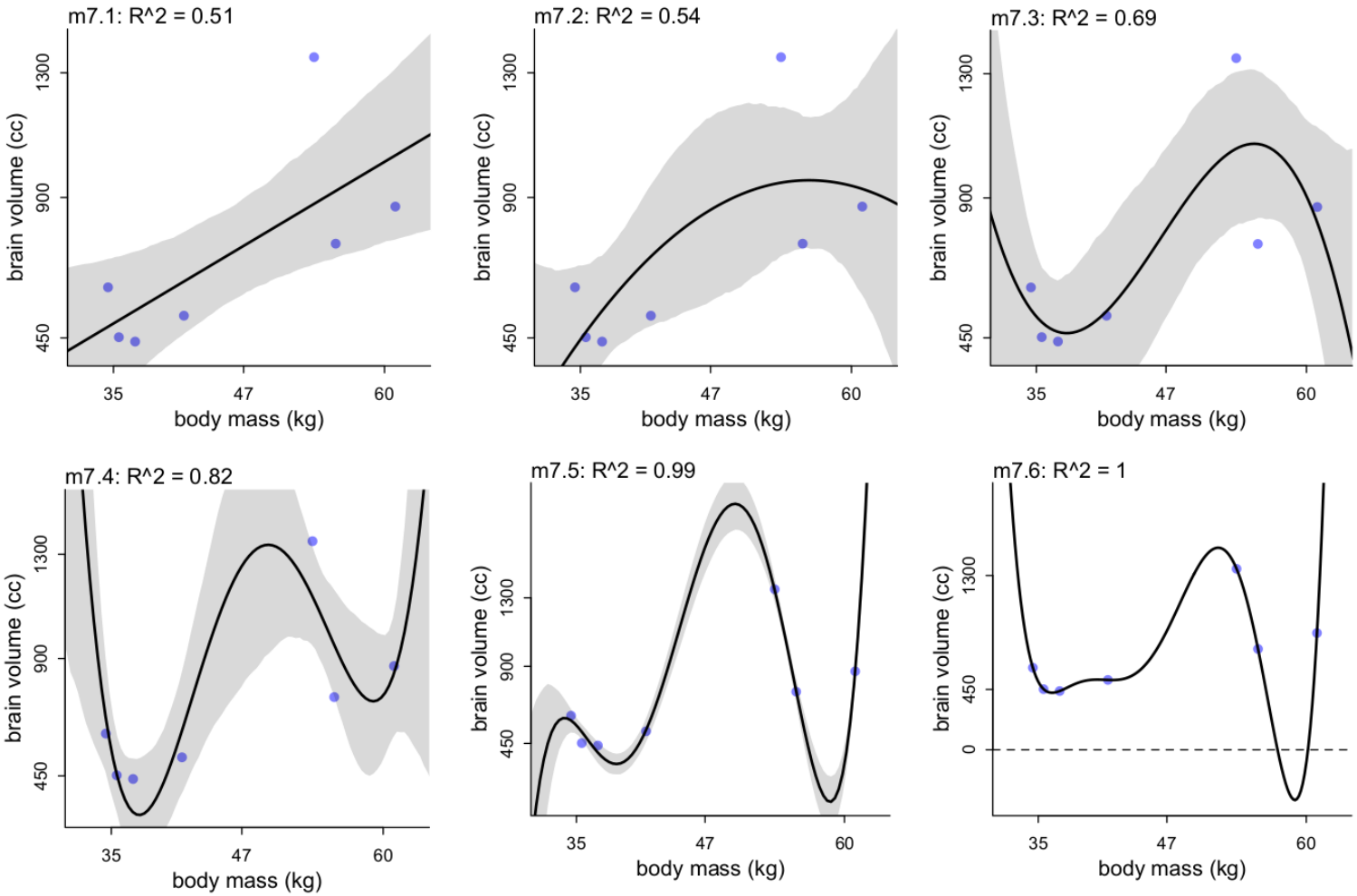
Slightly more complex model: a parabola

$$
\begin{aligned}
b_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha + \beta_1 m_i + \beta_2 m_i^2 \\
\alpha &\sim \text{Normal}(0.5, 1) \\
\beta_j &\sim \text{Normal}(0, 10),\ j = 1, 2 \\
\sigma &\sim \text{Log} - \text{Normal}(0, 1)
\end{aligned}
$$

Why not higher order polynomials?



Fit to the data: when R^2 becomes absurd



(SR figure 7.3)

Note: underfitting, i.e. being insensitive to the data also an issue.

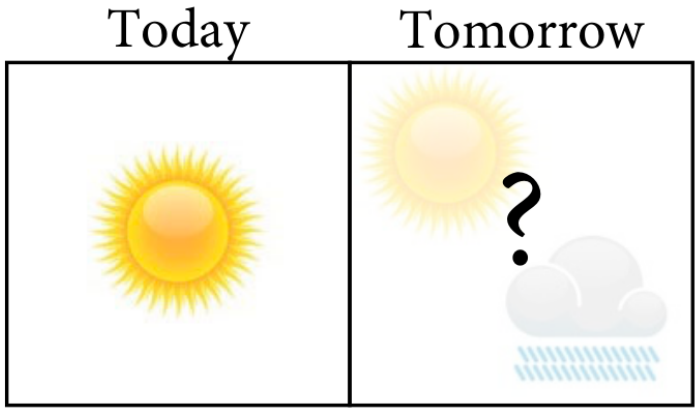
Going towards regularisation, CV and WAIC

- How can we measure accuracy?
- How can we measure the distance from the target?
- Can this distance be estimated from the data?
- Efficiently?
- Can we predict accuracy, i.e. the behaviour of the model, on new data?

- Let's talk about Information theory.

Information theory

- Machine prediction obeys *Information theory*.
- *Information*: reduction in uncertainty caused by learning an outcome.



The value of a prediction is context dependent

Cost benefit and accuracy in context



(see 7.2.1 Firing the weather person in SR book)

Information theory

- *Information*: reduction in uncertainty caused by learning an outcome.
- How to quantify *uncertainty*
  - continuous
  - increasing with the number of possible events
  - additive
- Good news: these criteria are intuitive and they lead to an effective solution: Shannon's information entropy

$$H(p) = -\mathbb{E}[\log(p_i)] = -\sum_{i=1}^{n} p_i \log(p_i)$$

The uncertainty in a probability distribution is minus the average log-probability of the events.

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Entropy to accuracy

- Imagine you have two probability distributions: p, the true (unknown, but let's pretend we can write it for now) and q the model.
- Question: how accurate is q for describing p?
- Key is in the *Divergence*: the additional uncertainty induced by using probabilities from one distribution (q) to describe another distribution (p). Often coined the Kullback-Leibler (KL) divergence:

$$D_{KL}(p, q) = \sum_i p_i(\log(p_i) - \log(q_i)) = \sum_i p_i \log\left(\frac{p_i}{q_i}\right)$$

  The KL divergence is the average difference of the log-probabilities of events described.
- The KL-divergence is not symmetrical (see Rethinking box: Divergence depends upon direction).

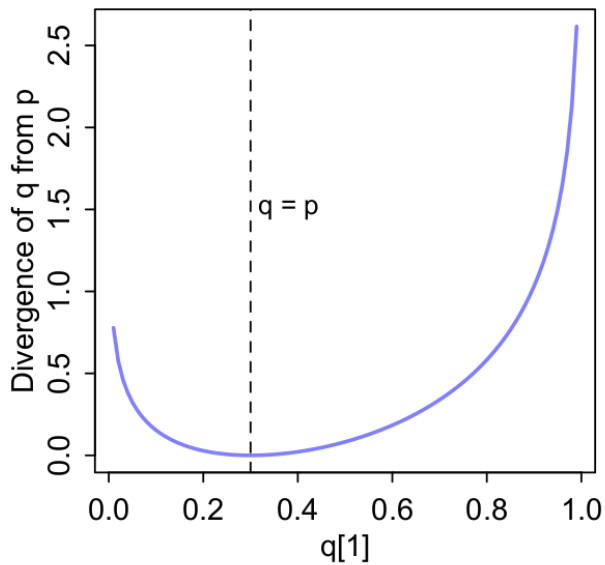
Computing divergence

$$D_{KL}(p,q) = \sum_i p_i(\log(p_i) - \log(q_i)) = \sum_i p_i \log\left(\frac{p_i}{q_i}\right)$$

```
p <- c(0.3, 0.7)

DKL <- function(p,q){
  sum(p*(log(p)-log(q)))}

q1seq <- seq(from = 0.01,
  to = 0.99, by = 0.01)
DKLseq <- sapply(q1seq, function(q1)
  DKL(p, c(q1, 1-q1)))
plot(q1seq, DKLseq)
```



The divergence is only 0 when $q = p$, otherwise > 0 and grows as q becomes more different to p.

Regularisation

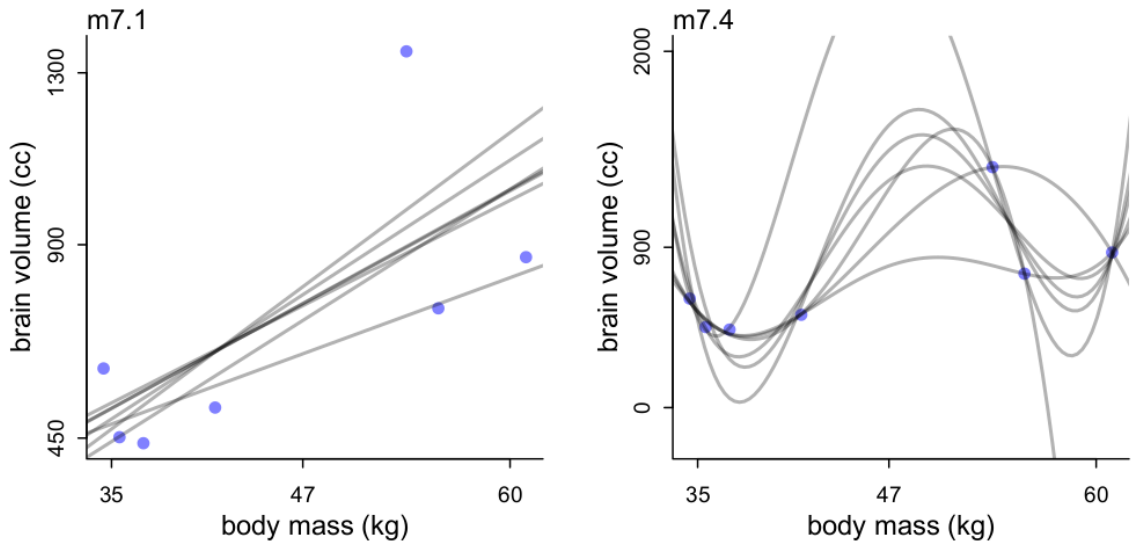
- Is rare in Science.
- Why?
- Probably because it is never taught.
- Makes “significant” results rarer.
- Most scientists do not judge on predictive accuracy.

CV and information criteria

- Can we estimate out-of-sample deviance?
- In theory: Cross-validation (see previous lecture, in particular for LOO-CV).
- In theory: Information criteria.
- In practice: both tend to perform similarly.

Cross-validation: a reminder

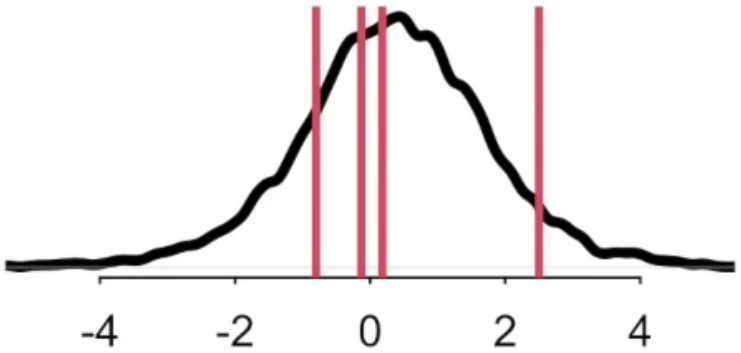
- Leave out some observations.
- Train the model on the remaining samples; score on those left out.
- Average over many left-out sets to get the out-of-sample (future) accuracy.



(SR book Figure 7.5)

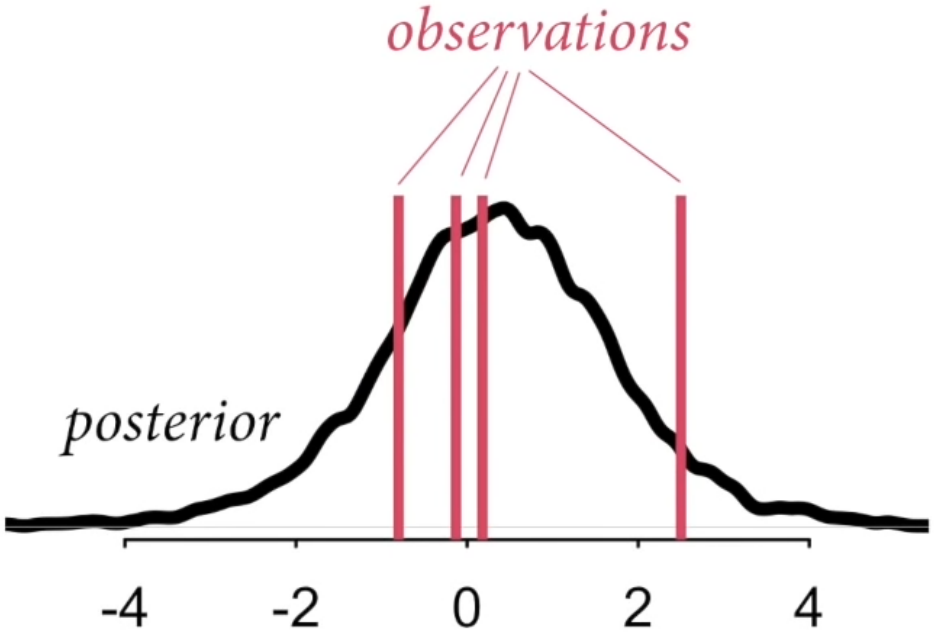
Importance sampling

- Importance sampling: use a single posterior distribution for N points to sample from each posterior from $N - 1$ points.
- Key idea: us the "importance" of each observation. Points with low probabilities has a strong influence on the posterior.
- This strong influence can be used to learn what the posterior would look like if it had been omitted.
- We can use pointwise-probabilities to reweigh samples from the posterior.



Importance sampling

A visualisation

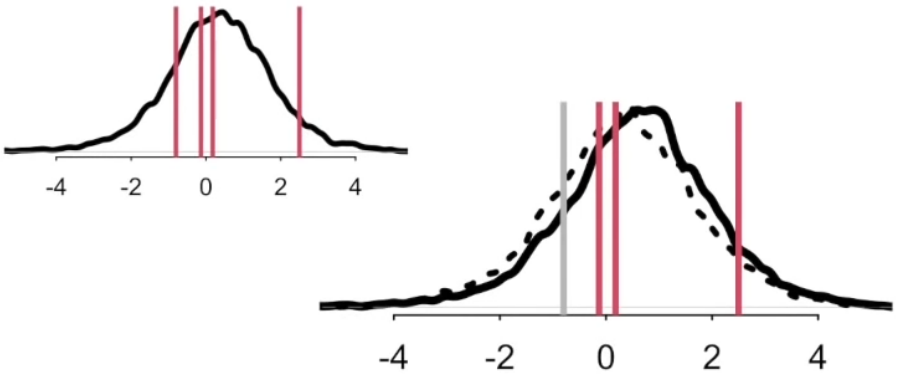


Each observation has contributed a different amount of information when building the posterior. Think about them one at a time.

Importance sampling

A visualisation

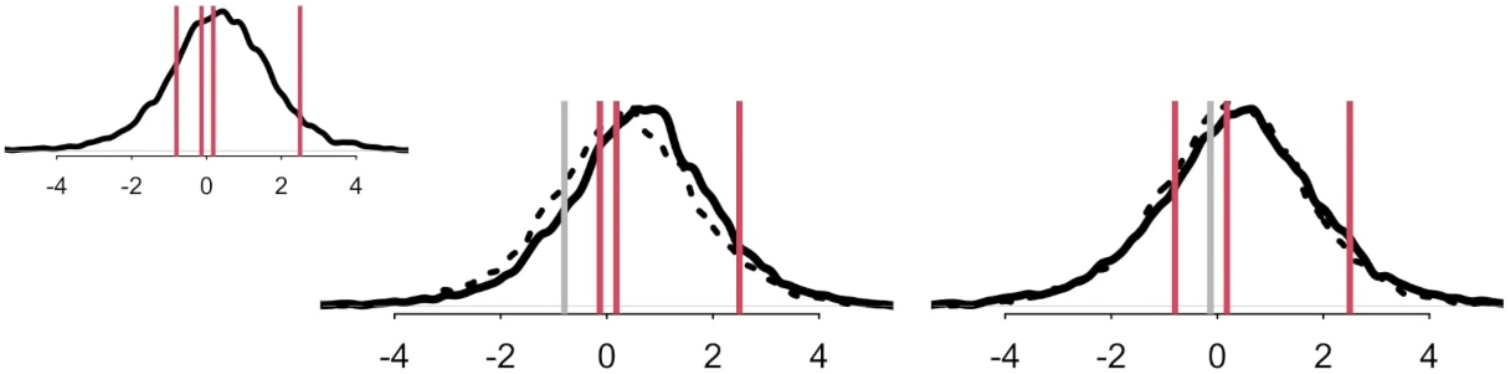
Imagine doing LOO-CV. Dashed = previously calculated posterior density. For the first point:



Importance sampling

A visualisation

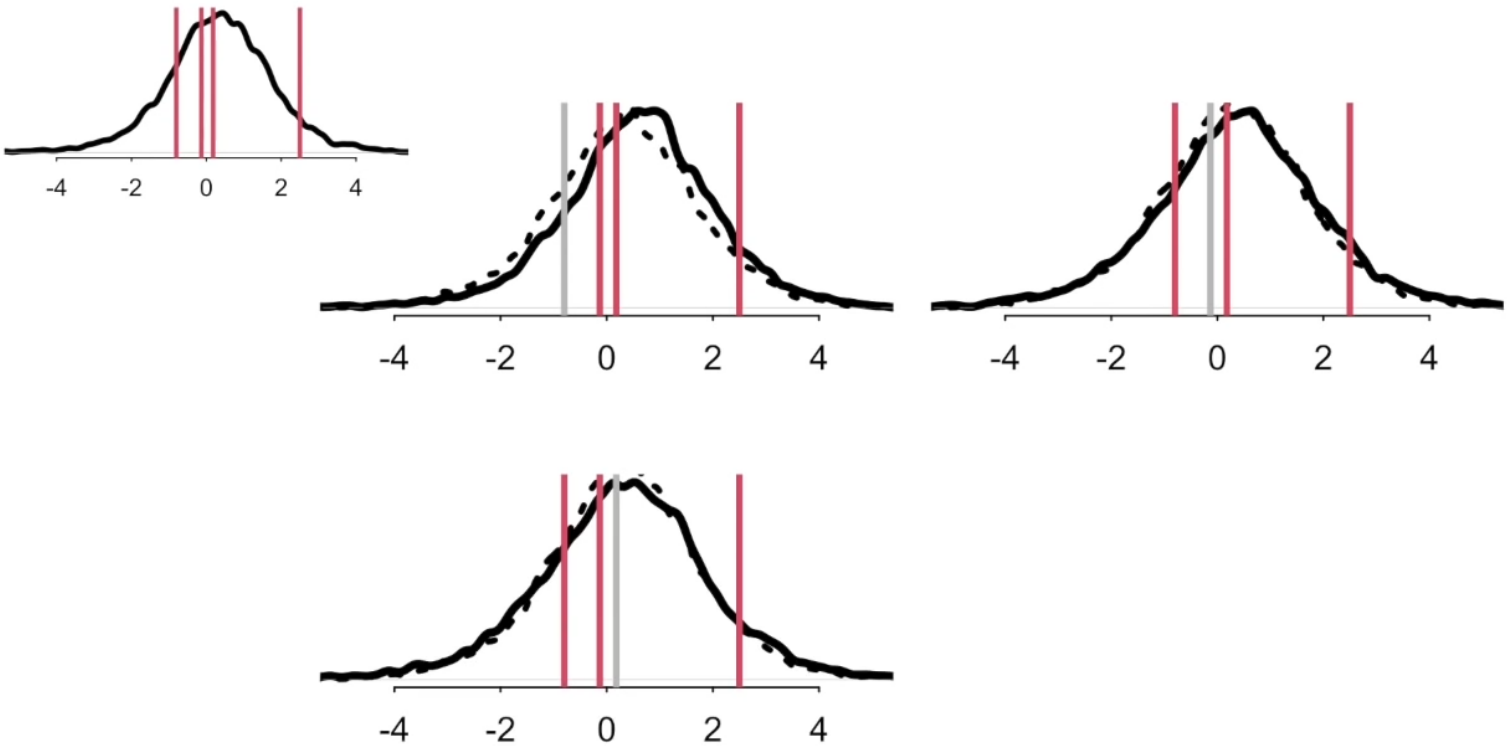
For the second point:



Importance sampling

A visualisation

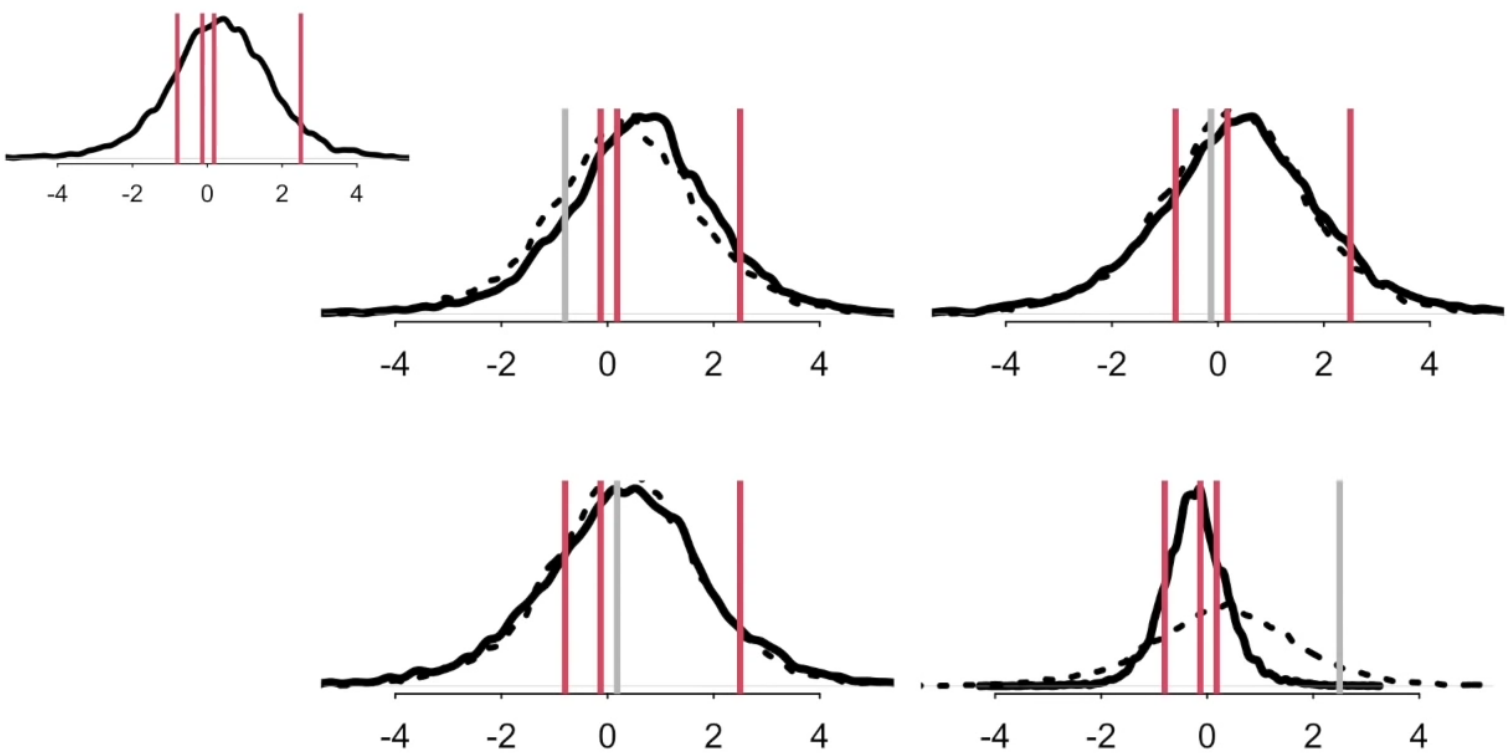
For the third point:



Importance sampling

A visualisation

The first 3 points seem to have little influence on the posterior when left out one at a time. Quite different for point 4:



Smooth CV

- Most common (and accurate): LOO-CV.
- Very expensive!
- Useful approximation: Importance sampling (IS).
- But tends to be unreliable with high variance (especially in small samples). It is "only" an estimator for the distribution, remember, we don't have it.
- More useful: Pareto-smoothed importance sampling (PSIS). It is more stable (has low variance).
- PSIS-LOO is accurate and has lots of useful diagnostics. It identifies important (high leverage) points ("outliers").
- `LOO()` function in the `rethinking` package. See also the `loo` package.

Akaike information criterion (AIC)

Another way to get validation score for a model

- Estimate information-theoretic measure of predictive accuracy (KL divergence)
- For flat priors and large samples:

$$\text{AIC} = (-2) \times \text{lppd} + 2k$$

  with lppd the log pointwise predictive density and k the number of parameters. $2k$ is the penalty term.
- Famous, but restrictive because of the assumptions (flat priors and large samples).
- Works OK for not too complex linear regressions with enough data. But we aim higher :)
- You may encounted DIC, OK with informative priors, but still assumes a(n approximate) multivariate Gaussian posterior and a sample size $N >>$ number of parameters k.

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Widely application information criterion (WAIC)

$$\text{WAIC}(y, \Theta) = -2 \left(\text{lppd} - \sum_i \text{var}_{Theta} log[p(y_i|\Theta)] \right)$$

- $\sum_i \text{var}_{Theta} log[p(y_i|\Theta)]$ is the penalty term, proportional to the variance in the posterior predictions. Sometimes called the *effective number of parameters* p_{WAIC}. The formula for WAIC is more complicated than that of AIC.
- It provides an approximation of the out-of-sample deviance that converges to the CV approximation in a large sample (can disagree in finite samples). It 'guesses out' the out-of-sample KL divergence.
- Each observation is penalised in a sense. What matters more is the relationship between the parameters, rather than their number. In multi-level modelling, adding parameters can reduce p_{WAIC}!
- Very similar to the PSIS score in practice. Both are pointwise. But no automatic diagnostics.

What's left?

- Tomorrow, we will compare the CV, PSIS and WAIC to see how well they approximate the out-of-sample accuracy.
- We see how to use them for model comparison.